Canonical Bell validity through two matrix inequalities

Abstract

A Bell inequality must hold for every state and both measurement angles. For the canonical Q_{222} slack, a polynomial change of state coordinates eliminates one angle: validity is equivalent to two 4×4 matrix inequalities with a 2×2 completion depending on the remaining angle. The equivalence includes zero source weights and projective endpoints. We then give a rational valid inequality for which no polynomial completion exists. A contact fixes its three possible polynomial parameters, and a two-coordinate quadratic test excludes them. A fixed algebraic completion or a common testing multiplier would be needed to obtain a finite global certificate by this construction.

1 Eliminating one measurement angle

Let $c = (d, \lambda, n, m, C, A, D, \alpha, \beta) \geq 0$ and put $D_u = 1 + u^2$, $D_v = 1 + v^2$. The established complete canonical parent is

$$\begin{aligned} F_c = {} & 4\lambda D_u (W + vX)^2 + 4nD_u D_v X^2 + 4m(vW - X + uvY - uZ)^2 \\ & + 4CD_v (uW - Y)^2 + 4AD_u D_v Y^2 + (4D + 2d)(uW + uvX - Y - vZ)^2 \\ & + (4\alpha + 2d) D_u (vY - Z)^2 + (4\beta + 2d) D_v (X + uZ)^2 - 2dD_u D_v Z^2. \end{aligned} \tag{1}$$

Define the symmetric 2×2 matrices

$$\begin{aligned} P(v) &= \begin{pmatrix} 4mv^2 & -4mv \\ -4mv & 4m + (4\beta + 2d) D_v \end{pmatrix}, \\ N(v) &= \begin{pmatrix} 4CD_v + 4D + 2d & (4D + 2d)v \\ (4D + 2d)v & (4D + 2d)v^2 \end{pmatrix}, \\ E(v) &= \begin{pmatrix} 4AD_v + (4\alpha + 2d)v^2 & -(4\alpha + 2d)v \\ -(4\alpha + 2d)v & 4\alpha - 2dv^2 \end{pmatrix}, \\ K(v) &= 4\lambda \begin{pmatrix} 1 & v \\ v & v^2 \end{pmatrix} + 4nD_v \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}. \end{aligned} \tag{2}$$

All entries are affine in c and quadratic in the testing variable v. For $a = (S, T)^T$, $b = (P_0, Q_0)^T$, and $U = u^2$, make the polynomial change

$$W = S + UP_0, \quad X = T + UQ_0, \quad Y = u(S - P_0), \quad Z = u(T - Q_0). \tag{3}$$

The state coordinates P_0, Q_0 are unrelated to the matrix $P(v)$. Direct substitution gives the exact identity

$$F_c = D_u \Phi_c(U, v; a, b), \qquad \boxed{\begin{aligned} \Phi_c = {} & (1 + U) a^T P a + U(1 + U) b^T N b \\ & + U(a - b)^T E (a - b) + (a + Ub)^T K (a + Ub). \end{aligned}} \tag{4}$$

For example, the m, C, D, α, β event roots in (1) become respectively $D_u(vS - T)$, $uD_u P_0$, $uD_u(P_0 + vQ_0)$, $u\{v(S - P_0) - (T - Q_0)\}$, and $D_u T$. These substitutions prove (4) without a source-dependent inverse.

Write $\psi = (W, X, Y, Z)^T$ and let $\mathcal{P}_c(U, v)$ be the symmetric matrix of Φ_c in (a, b). Its three coefficients are

$$\mathcal{P}_c = \begin{pmatrix} P+K & 0 \\ 0 & 0 \end{pmatrix} + U \begin{pmatrix} P+E & K-E \\ K-E & N+E \end{pmatrix} + U^2 \begin{pmatrix} 0 & 0 \\ 0 & N+K \end{pmatrix}. \tag{5}$$

The change matrix has determinant $u^2 D_u^2$. Its polynomial inverse numerator is

$$L(u) = \begin{pmatrix} u & 0 & u^2 & 0 \\ 0 & u & 0 & u^2 \\ u & 0 & -1 & 0 \\ 0 & u & 0 & -1 \end{pmatrix}, \qquad \boxed{u^2 D_u F_c(\psi) = (L(u)\psi)^T \mathcal{P}_c(u^2, v)(L(u)\psi).} \tag{6}$$

To turn (6) into squares, we first factor its quadratic dependence on U.

2 A pointwise matrix completion

Theorem 1. *For any fixed real source c and fixed real Bob test v, the following are equivalent:*

1. *$\mathcal{P}_c(U, v) \succeq 0$ for every $U \geq 0$.*
2. *There exists a real 2×2 matrix J such that*

$$\boxed{\mathcal{G}_J = \begin{pmatrix} P+K & J \\ J^T & N+K \end{pmatrix} \succeq 0, \qquad \mathcal{H}_J = \begin{pmatrix} P+E & K-E-J \\ K-E-J^T & N+E \end{pmatrix} \succeq 0.} \tag{7}$$

The equivalence also holds over every real closed field. In particular, complete canonical validity is exactly

$$\boxed{\forall v \in \mathbb{R}\ \exists J \in \mathbb{R}^{2\times 2} : \mathcal{G}_J(v) \succeq 0, \quad \mathcal{H}_J(v) \succeq 0.} \tag{8}$$

All source faces, including $d = 0$ and arbitrary zero event weights, are retained.

Proof. The reverse implication is the explicit identity

$$\Phi_c = \begin{pmatrix} a \\ Ub \end{pmatrix}^T \mathcal{G}_J \begin{pmatrix} a \\ Ub \end{pmatrix} + U \begin{pmatrix} a \\ b \end{pmatrix}^T \mathcal{H}_J \begin{pmatrix} a \\ b \end{pmatrix}. \tag{9}$$

For the forward implication, $\mathcal{P}_c(t^2, v)$ is a positive semidefinite univariate matrix polynomial of degree at most four. The real-line matrix Fejér–Riesz theorem gives a polynomial matrix factor of degree at most two; real and imaginary parts give a real sum of matrix squares [1, Theorem 1.2]. Averaging its values at t and $-t$ separates even and odd powers. Consequently

$$\mathcal{P}_c(U, v) = (B_0 + UB_2)^T (B_0 + UB_2) + UB_1^T B_1$$

for rectangular real matrices B_0, B_1, B_2. The constant coefficient in (5) has its last two rows and columns zero, so $B_0 = [R \ \ 0]$. The coefficient of U^2 has its first two rows and columns zero, so $B_2 = [0 \ \ T]$. This uses only the elementary fact that a zero diagonal entry of a positive semidefinite Gram matrix forces its entire row and column to vanish. Setting $J = R^T T$ gives $\mathcal{G}_J = [R\ T]^T [R\ T]$ and $\mathcal{H}_J = B_1^T B_1$, proving (7).

This is a fixed finite-dimensional first-order assertion in the real closed field language: positivity is expressed by principal minors, and the half-line condition by quantification over U and a four-vector. Transfer therefore proves precisely the same assertion over every real closed field.

For $u \neq 0$, (3) is invertible and so positivity of F_c is equivalent to positivity of $\mathcal{P}_c(u^2, v)$. The excluded value $u = 0$ is recovered by polynomial continuity; all $U > 0$ are attained. More explicitly, (1) has bidegree at most $(2, 2)$ in (u, v), so

$$F_c^{\mathrm{h}}(r, s; \psi) = r_0^2 s_0^2 F_c(r_1/r_0, s_1/s_0; \psi)$$

is polynomial in the homogeneous angle pairs $r = (r_0, r_1)$ and $s = (s_0, s_1)$. On their unit circles it equals the normalization $F_c/(D_u D_v)$. Continuity recovers the Alice and Bob projective endpoints. Over a real closed field the same endpoint statement follows by the sign of the first nonzero coefficient in the corresponding local polynomial chart. No source coefficient was divided out. □

There is also an elementary check on the face $d = 0$: then $E \succeq 0$, and $J = K$ makes both matrices in (7) sums of the positive diagonal blocks P, N and a square block built from K or E. More generally adding a nonnegative event source with $d = 0$ preserves any certificate by adding its K to J. The unresolved completion is therefore associated with the signed core, not with adding positive events.

The remaining angle enters the completion through $J(v)$. A finite global certificate would require these matrices to lie in one fixed function space. Pointwise transfer permits algebraic dependence on both the source and the angle; it supplies no such space in the tensor product of their independent fields. We therefore examine the simplest possibility, polynomial $J(v)$.

3 Why polynomial completions do not suffice

Proposition 2. *If polynomial entries $J(v)$ satisfy (7) for every real v, they have degree at most two. A selector can then be chosen of the form*

$$J(v) = \begin{pmatrix} p_0 + p_2 v^2 & q_1 v \\ r_1 v & s_0 + s_2 v^2 \end{pmatrix}.$$

On the entire C1 face $\lambda = n = 0$, existence of any polynomial selector is equivalent to existence of one with just three scalar parameters:

$$\boxed{J(v) = \begin{pmatrix} pv^2 & 0 \\ rv & sv^2 \end{pmatrix}.} \tag{10}$$

Proof. Every 2×2 principal minor of $\mathcal{G}_J$ gives $|J_{ij}(v)|^2 \leq (P + K)_{ii}(N + K)_{jj}$. The right side has degree at most four, proving the degree bound, including cases where it is identically zero. With $T = \mathrm{diag}(1, -1)$, each matrix in (2) obeys $M(-v) = TM(v)T$. Replace $J(v)$ by $\frac{1}{2}\{J(v) + TJ(-v)T\}$; convexity of the positive semidefinite cone retains both inequalities and gives the displayed parity.

On C1, $P(0) = \mathrm{diag}(0, 4m + 4\beta + 2d)$ and $N(0) = \mathrm{diag}(4C + 4D + 2d, 0)$. Zero diagonal entries in $\mathcal{G}_J(0)$ force $p_0 = s_0 = 0$. Finally

$$q_1^2 v^2 = |J_{12}(v)|^2 \leq 4m(4D + 2d)v^4$$

for every v, so $q_1 = 0$. This proof does not assume any of the source weights is positive. □

A small rational example already excludes polynomial completions. Set $\lambda = n = 0$ and

$$(d, m, C, A, D, \alpha, \beta) = \left(359, \frac{75}{16}, 60, 416, \frac{41}{2}, \frac{831}{2}, \frac{151}{2}\right).$$

Let $P_*(u,v)$ be the four-dimensional state matrix of (1) at this source. Its validity follows from four elementary AM–GM inequalities. Put $\xi = 4u^2$, $\eta = v^2/4$, and

$$\begin{aligned} M(\xi,\eta) = {} & 3757645(\xi + \xi^{-1} - 2) + 14022297(\eta + \eta^{-1} - 2) \\ & + 3732435(\eta/\xi + \xi + \eta^{-1} - 3) \\ & + 8179670(\xi\eta + \xi^{-1} + \eta^{-1} - 3). \end{aligned}$$

Each bracket is nonnegative for $\xi, \eta > 0$: its positive terms have product one. Direct expansion gives

$$\det P_*(u,v) = 28800 D_u^2 D_v^2 u^2 v^2 M(\xi,\eta). \tag{11}$$

The $Z = 0$ principal block B_* is strictly positive when $u \neq 0$. Indeed, the positive m, C, A terms of (1) give

$$B_* \succeq T^T \operatorname{diag}(4m, 4CD_v, 4AD_uD_v)T, \qquad T = \begin{pmatrix} v & -1 & uv \\ u & 0 & -1 \\ 0 & 0 & 1 \end{pmatrix}, \quad \det T = u.$$

All remaining terms on $Z = 0$ are nonnegative. The scalar Schur complement therefore proves $P_* \succeq 0$ for $uv \neq 0$ from (11). Polynomial continuity supplies the finite axes, and the homogeneous angle charts supply the projective endpoints. Separating real and imaginary parts covers complex states as well.

The contact has $u_* = 1/2$, $v_* = 2$ and $\psi_* = (2, -1/4, 1/4, 1)^T$. Under (3) it becomes

$$U_* = \frac{1}{4}, \qquad a_* = \begin{pmatrix} 17/10 \\ 1/5 \end{pmatrix}, \qquad b_* = \begin{pmatrix} 6/5 \\ -9/5 \end{pmatrix}. \tag{12}$$

Substitution verifies the full state equation $\mathcal{P}_c(U_*, v_*)(a_*, b_*)^T = 0$.

Theorem 3. *This valid source has no polynomial selector satisfying* (7) *for all* v*. Equivalently, its polynomial* $\Phi_c(u^2, v; a, b)$ *is not a sum of polynomial squares in* (u, v, a, b)*.*

Proof. By Proposition 2, a polynomial selector would have the form (10). At the contact, both nonnegative terms in (9) vanish. Therefore

$$J(v_*)b_* = -P(v_*)a_*/U_*, \qquad J(v_*)^T a_* = -U_* N(v_*) b_*. \tag{13}$$

The first coordinate of the first equation fixes p, and the second coordinate of the second fixes s. Either remaining coordinate then gives r. With (12), the unique solution is

$$p = -100, \qquad r = 2000, \qquad s = 1200. \tag{14}$$

At $v = 1$, the principal submatrix of $\mathcal{G}_J$ on (a_2, b_1) would consequently be

$$\begin{pmatrix} 8235/4 & 2000 \\ 2000 & 1280 \end{pmatrix}.$$

Its quadratic value on $(1, -1)$ is $-2645/4$, contradicting $\mathcal{G}_J(1) \succeq 0$.

Finally, a polynomial sum of squares for $\Phi_c(u^2, v; a, b)$ can use roots of degree at most two in u, at most one in v, and homogeneous linear degree in the four state coordinates: higher terms cannot cancel in a sum of real squares. Averaging in $u \mapsto -u$ and using the constant and leading state supports exactly as in Theorem 1 would produce a polynomial selector. Conversely a polynomial selector gives polynomial matrix factorizations of its two positive univariate matrices by Fejér–Riesz and then a polynomial sum of squares through (9). This proves the equivalence. □

4 Polynomial multipliers and half-line completions

The endpoint constraint on J_{12} distinguishes polynomials from algebraic functions. An odd algebraic function such as $v|v|$ can vanish quadratically at zero without vanishing identically. This extra freedom suffices for the source in Theorem 3: the half-line construction gives an explicit rational quadratic selector on $v \geq 0$, extended to $v < 0$ by reflection. Its four constant Gram blocks have orders $8, 8, 4, 4$. Adding source nonnegativity gives an affine pencil of total order 33, with 44 auxiliaries. Its projection is closed.

A polynomial multiplier would instead require one fixed nonzero nonnegative $q(v)$ such that every valid source admits a polynomial $Z_c(v)$ satisfying

$$\begin{pmatrix} q(P+K) & Z_c \\ Z_c^T & q(N+K) \end{pmatrix} \succeq 0, \qquad \begin{pmatrix} q(P+E) & q(K-E)-Z_c \\ q(K-E)-Z_c^T & q(N+E) \end{pmatrix} \succeq 0 \quad (v \in \mathbb{R}).$$

The first inequality forces $\deg Z_c \leq \deg q + 2$. Univariate matrix factorization would then give a fixed finite testing family for $q(v)u^2 D_u F_c$, with coefficient equations affine in the source and Gram matrices. Since the multiplier is positive on a dense set, its certificate would also imply validity of F_c at every angle by continuity. The fixed-multiplier theorem rules this out for every such q. It supplies a strictly valid rational source family whose rescaled full-state slack tends to the Motzkin polynomial plus a constant. Every fixed separable polynomial multiplier leaves a square monomial in this limit, which can be cancelled from a hypothetical polynomial sum of squares to obtain a contradiction.

A square-root factor on the half-line removes that limiting obstruction. The remaining question is whether every valid canonical source admits a quadratic half-line selector. The pointwise equivalence (8) and the explicit half-line cone express this question as a simultaneous matrix-completion problem.

References

[1] A. Zalar, *Matrix Fejér–Riesz theorem with gaps*, Theorem 1.2, arXiv:1503.06034; author's full text.